

HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG
KHOA CÔNG NGHỆ THÔNG TIN



MAJOR ASSIGNMENT
SUBJECT NAME: INTRODUCTION TO
SOFTWARE ENGINEERING

CLASS NAME: E23CN5

PROJECT NAME:
National Library Management System

PROJECT CODE: 3

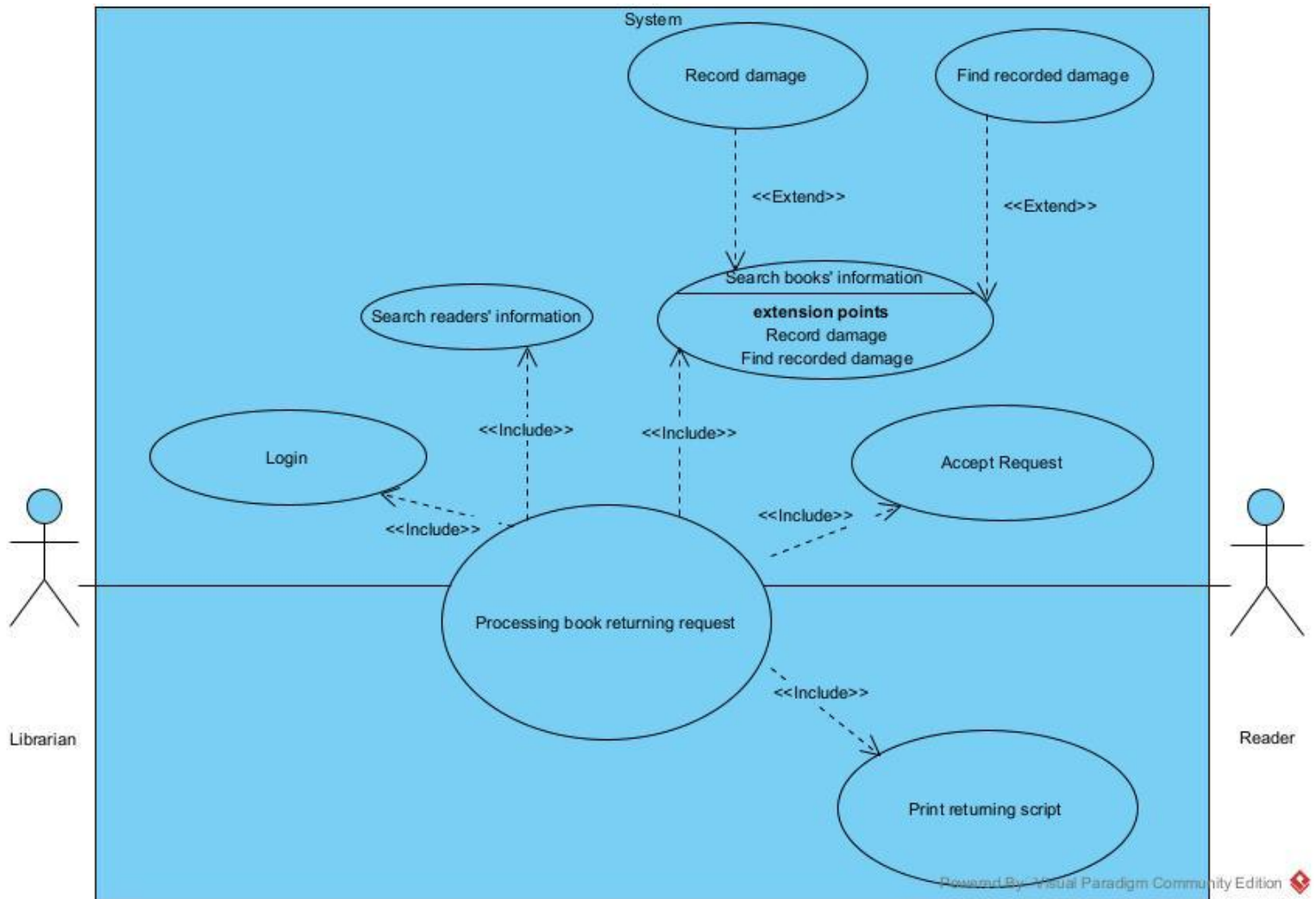
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Module: Returning of books

Weekly Requirement: Summary Report

1. Detail UC diagram for the “Returning book” module.



The Login use case allows the librarian to log into the system and execute the Book Return use case.

The Search Readers' Information use case allows the librarian to search for information about readers returning books.

The Search Books' Information use case allows the librarian to search for information about books brought in for return.

The Record Damage use case allows the librarian to record each damaged type for each book.

The Find Recorded Damage use case allows the librarian to search for damage to each book.

The Print Returning Script use case allows the librarian to print the reader's book return receipt.

The Accept Request use case allows the librarian to save the book returning receipt to the database.

2. Standard Scenario of the module

Scenario	Returning books																																																																																
Actors	Librarian, Reader																																																																																
Pre-condition	The librarian has an account of librarian, the reader has a reader card and has borrowed some books before.																																																																																
Post-condition	The reader has returned some books																																																																																
Main events	1. Librarian A logs into the login page with username=a, password=abc123, to handle reader B’s returning book requests at the library counter. 2. The system displays the librarian's home page having the option to return books. 3. Librarian chooses the returning book option. 4. The system displays the reader search interface. - Search bar: Input for searching - Button: Search Reader, Scan Reader Card Search Reader Search Scan Card 5. The librarian selects the Scan Reader Card button and then uses a scanner connected to the system to scan the reader card barcode or enter Reader name and selects Search Reader. 6. The system displays the reader information interface. -Button: Back, Scan Book, Next - Reader Information <table><tr><td>Reader ID</td><td>Name</td><td>DoB</td><td>Address</td><td>Phone number</td><td>Bar Code</td></tr><tr><td>001</td><td>B</td><td>01/01/2026</td><td>Hà Nội</td><td>0123456789</td><td>0123456789</td></tr></table> - List of borrowed books that have not been returned <table><tr><td>Order</td><td>Book Code</td><td>Bar Code</td><td>Name</td><td>Author</td><td>Borrowing Date</td><td>Due Date</td><td>Cover Price</td></tr><tr><td>1</td><td>00001</td><td>12345</td><td>Harry Potter</td><td>J.K Rowling</td><td>11/03/2026</td><td>11/04/2026</td><td>10000vnd</td></tr><tr><td>2</td><td>00002</td><td>12346</td><td>Diary of a Cricket</td><td>Tô Hoài</td><td>11/03/2026</td><td>11/04/2026</td><td>20000vnd</td></tr><tr><td colspan="5">The total books:</td><td>2</td><td colspan="2"></td></tr></table> - List of borrowed books that have been returned <table><tr><td>Order</td><td>Book Code</td><td>Bar Code</td><td>Name</td><td>Author</td><td>Cover Price</td><td>Due Date</td><td>Returning Date</td><td>Total Fine Amount</td></tr><tr><td>1</td><td>00003</td><td>12347</td><td>Tấm Cám</td><td>Unknown</td><td>30000vnd</td><td>01/04/2026</td><td>20/03/2026</td><td>0vnd</td></tr><tr><td>2</td><td>00004</td><td>12348</td><td>The lotus shoes</td><td>Jane Yang</td><td>40000vnd</td><td>01/04/2026</td><td>21/03/2026</td><td>40000vnd</td></tr><tr><td colspan="5">The total of returned books</td><td>2</td><td colspan="3"></td></tr></table> Scan book Back Next 7. The librarian asks the reader if they want to return any books.	Reader ID	Name	DoB	Address	Phone number	Bar Code	001	B	01/01/2026	Hà Nội	0123456789	0123456789	Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover Price	1	00001	12345	Harry Potter	J.K Rowling	11/03/2026	11/04/2026	10000vnd	2	00002	12346	Diary of a Cricket	Tô Hoài	11/03/2026	11/04/2026	20000vnd	The total books:					2			Order	Book Code	Bar Code	Name	Author	Cover Price	Due Date	Returning Date	Total Fine Amount	1	00003	12347	Tấm Cám	Unknown	30000vnd	01/04/2026	20/03/2026	0vnd	2	00004	12348	The lotus shoes	Jane Yang	40000vnd	01/04/2026	21/03/2026	40000vnd	The total of returned books					2			
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The total of returned books					2																																																																												

8. The reader give the librarian a copy of the book "Diary of a Cricket".

(Repeat steps 9 to 10 until the librarian scans all the books.)

9. The librarian clicks the "Scan book" button and used a scanner connected to the system to scan the barcode of the book.

10. The system updates the reader information interface:

-Button: Back, Scan Book, Next

- Reader Information

Reader ID	Name	DoB	Address	Phone number	Bar Code
001	B	01/01/2026	Hà Nội	0123456789	0123456789

- List of borrowed books that have not been returned

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover price
1	00001	12345	Harry Potter	J.K Rowling	11/03/2026	11/04/2026	10000vnd
The total books:					1		

- List of scanned books

Order	Book Code	Bar Code	Name	Author	Due Date	Returning Date	Cover price	Damage Status
1	00002	12346	Diary of a Cricket	Tô Hoài	11/04/2026	12/04/2026	20000vnd	TORN
The total books:					1			

- List of borrowed books that have been returned

Order	Book Code	Bar Code	Name	Author	Cover price	Due Date	Returning Date	Total Fine Amount
1	00003	12347	Tâm Cám	Unknown	30000vnd	01/04/2026	20/03/2026	0vnd
2	00004	12348	The lotus shoes	Jane Yang	40000vnd	01/04/2026	21/03/2026	40000vnd
The total of returned books					2			

Scan book

Back

Next

11. The staff member checks books, finds no problems, and presses the "Next" button.

12. The system switches to the returning receipt interface.

- Reader Information:

Reader ID	Name	Bar Code	Script Code
001	B	0123456789	1234561234

- List of borrowed books that have not been returned

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover price
1	00001	12345	Harry Potter	J.K Rowling	11/03/2026	11/04/2026	10000vnd

The total books: 1

- List of books fined for late returning

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Returning Date	Cover price	Fine Amount (20%)
1	00002	12346	Diary of a Cricket	Tô Hoài	11/03/2026	11/04/2026	12/04/2026	20000vnd	4000vnd
The total					4000vnd				

- List of books fined for returning damaged book

Order	Book Code	Bar Code	Name	Author	Cover price	Damage Status Now	Fine Amount
1	00002	12346	Diary of a Cricket	Tô Hoài	20000vnd	TORN	20000vnd
The total					20000vnd		

TOTAL FINE: 24000vnd DEPOSIT: 20000vnd

Submit

Back to Main Page

Back

13. The librarian let the reader check their book returning receipts.
14. The reader confirms that it's correct
15. The librarian handle the payment and click Submit button
16. The system stores information about book returns, notifies that the book return was successful, and prints out the book return receipt.

Exception

1. The system shows wrong login information input
 - 1.1 The librarian clicks OK
 - 1.2 The system return to login page with username = a, password = abc
 - 1.3 The librarian changes password = abc123 and clicks LOGIN
 - 1.4 The system shows homepage
6. The system reports that the reader information scan was unsuccessful.
 - 6.3 The librarian clicks the OK button on the notification box.
 - 6.2 The system now comes back to the reader search interface.
 - Search bar: Input for searching
 - Button: Search Reader, Scan Reader Card
 - 6.4 The librarian asks about the reader' name
 - 6.5 The reader tells that his/her name is "B"
 - 6.6 The librarian enters "B" in search bar and clicks Search Reader button
 - 6.7 The system shows the line
ID=001, Name=B, DoB = 01/01/2026, Address=Hà Nội, Phone number=0123456789
 - Search bar: Input for Reader ID
 - Button: Search Reader, Scan Reader Card

Order	Reader ID	Name	DoB	Address	Phone number	Bar Code
1	001	B	01/01/2001	Hà Nội	0123456789	0123456789
2	002	Bình	12/02/2003	Hà Nội	1214142423	2334455667

- 6.8 The librarian clicks on the first row
- 6.9 The system displays the reader information interface.
 - Button: Back, Scan Book, Next
 - Reader Information

Reader ID	Name	DoB	Address	Phone number	Bar Code
001	B	01/01/2026	Hà Nội	0123456789	0123456789

- List of borrowed books that have not been returned

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover price
1	00001	12345	Harry Potter	J.K Rowling	11/03/2026	11/04/2026	10000vnd
2	00002	12346	Diary of a Cricket	Tô Hoài	11/03/2026	11/04/2026	20000vnd
The total books:					2		

- List of borrowed books that have been returned

Order	Book Code	Bar Code	Name	Author	Cover price	Due Date	Returning Date	Total Fine Amount
1	00003	12347	Tâm Cầm	Unknown	30000vnd	01/04/2026	20/03/2026	0vnd
2	00004	12348	The lotus shoes	Jane Yang	40000vnd	01/04/2026	21/03/2026	40000vnd
The total of returned books					2			

Scan book

Back

Next

10. The system notifies that the current book is not on the list of unreturned borrowed items.

10.1 The librarian informed the reader that the book they brought in was not one of the books they had borrowed and returned the book to them.

10.2 The system displays the reader information interface.

-Button: Back, Scan Book, Next

- Reader Information

Reader ID	Name	DoB	Address	Phone number	Bar Code
001	B	01/01/2026	Hà Nội	0123456789	0123456789

- List of borrowed books that have not been returned

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover price
1	00001	12345	Harry Potter	J.K Rowling	11/03/2026	11/04/2026	10000vnd
2	00002	12346	Diary of a Cricket	Tô Hoài	11/03/2026	11/04/2026	20000vnd
The total books:					2		

- List of borrowed books that have been returned

Order	Book Code	Bar Code	Name	Author	Cover price	Due Date	Returning Date	Total Fine Amount
1	00003	12347	Tấm Cám	Unknown	30000vnd	01/04/2026	20/03/2026	0vnd
2	00004	12348	The lotus shoes	Jane Yang	40000vnd	01/04/2026	21/03/2026	40000vnd
The total of returned books					2			

Scan book

Back

Next

11. The librarian noticed a page was torn in the book and clicked on the button in the “Damage Status Now” column in the list of scanner books.

11.1 The system now displays an interface for selecting damage state of the book:

- Book Code: 000002

- BarCode: 12346

- Name: Diary of a Cricket

- Author: Tô Hoài

- Borrowing Date: 11/03/2026

- Due Date: 11/04/2026

- Returning Date: 12/04/2026

- Cover Price: 20000vnd

- Type of Damage:

Type	Fine Amount (Based on cover price)	Select
Torn	100%	☒
Graffiti	50%	☐
Mold	100%	☐

Crumpled	100%	☐
Bent	80%	☐
Add		

- “Back” button
- “Accept” button

11.2 The librarian double clicks on the “Torn” row to update the Fine Amount, click on the Button of the column Select to select

11.3 The system now displays the updated book information interface:

- Book Code: 000002
- BarCode: 12346
- Name: Diary of a Cricket
- Author: Tô Hoài
- Borrowing Date: 11/03/2026
- Due Date: 11/04/2026
- Returning Date: 12/04/2026
- Cover Price 20000vnd
- Type of Damage:

Type	Fine Amount	Select
Torn	100%	☒
Graffiti	50%	☐
Mold	100%	☐
Crumpled	100%	☐
Bent	80%	☐
Add		

- “Back” button
- “Accept” button

11.4 The librarian click on Accept button.

11. The librarian noticed a stained page in the book and clicked on the button in the “Damage Status Now” column in the list of scanner books.

11.1 The system now displays an interface for selecting damage state of the book:

- Book Code: 000002
- BarCode: 12346
- Name: Diary of a Cricket
- Author: Tô Hoài
- Borrowing Date: 11/03/2026
- Due Date: 11/04/2026
- Returning Date: 12/04/2026
- Cover Price: 20000vnd
- Type of Damage:

Type	Fine Amount	Select
Torn	100%	☐
Graffiti	50%	☐
Mold	100%	☐
Crumpled	100%	☐
Bent	80%	☐
Add		

- “Back” button
- “Accept” button

11.2 The librarian click on “Add” row to add new type of Damage, enter fine amount and click Select button

- Book Code: 000002

- BarCode: 12346
- Name: Diary of a Cricket
- Author: Tô Hoài
- Borrowing Date: 11/03/2026
- Due Date: 11/04/2026
- Returning Date: 12/04/2026
- Cover Price: 20000vnd
- Type of Damage:

Type	Fine Amount	Select
Torn	100%	☐
Graffiti	50%	☐
Mold	100%	☐
Crumpled	100%	☐
Bent	80%	☐
Stained	50%	☒
Add		

- “Back” button
- “Accept” button

11.3 The librarian clicks “Accept” button

14. The reader told the librarian that they want to return another copy of the book "Harry Potter".

14.1 The librarian clicks “Back” button

14.2 The system displays the reader information interface.

- Button: Back, Scan Book, Next
- Reader Information

Reader ID	Name	DoB	Address	Phone number	Bar Code
001	B	01/01/2026	Hà Nội	0123456789	0123456789

- List of borrowed books that have not been returned

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover price
1	00001	12345	Harry Potter	J.K Rowling	11/03/2026	11/04/2026	10000vnd
The total books:					1		

- List of borrowed books that have been returned

Order	Book Code	Bar Code	Name	Author	Cover price	Due Date	Returning Date	Total Fine Amount
1	00003	12347	Tám Cám	Unknown	30000vnd	01/04/2026	20/03/2026	0vnd
2	00004	12348	The lotus shoes	Jane Yang	40000vnd	01/04/2026	21/03/2026	40000vnd
The total of returned books					2			

Scan book

Back

Next

14.3 The client scans the book and clicks Next to continue to the payment.

3. Entity class diagram of the Analysis phase

Entity class extraction using noun extraction method

S1: Describe the system in a short paragraph

The Book Return module facilitates the book return process in a library.

In this module, readers bring books to the librarian's desk. Readers present their library card to the librarian. The librarian receives the card and scans it to obtain reader information.

The reader information is displayed (including ID, name, date of birth, address, phone number, and barcode) + a list of borrowed books that have not been returned + a list of borrowed books that have been returned. The librarian can scan each book individually. During the scanning process, if the librarian discovers a damaged or late-returned book, they will record the error on a penalty slip to select type of damage for each book later.

After the book scanning process is complete, the librarian will process the payment for the reader.

If the list of unreturned borrowed books by readers is still available, the librarian will announce it and print a borrowing script including the reader's ID, name, barcode, borrowing script barcode, and a list of remaining borrowed books, each book on a separate line: ID, book title, author, barcode, borrowing date, due date.

S2 + S3: Extract all name and evaluate extracted noun

- Librarian => User: Username, Password, Fullname, Role
- Reader => Reader: Name, DateOfBirth, PhoneNumber, Barcode, Address
- Book => Book: Name, Code, Author, PublicationYear, Price, Description, Barcode.
- Borrowing receipt => Borrowing receipt: Barcode, DepositAmount, Note, CreatedDate
- Returning receipt => Returning receipt: Barcode, Note, CreatedDate
- Damage => Damage: Name, FineRate
- Code, Name, Date of Birth, Address, Phone Number, Borrowing Receipt Barcode, Book Title, Author, Borrowing Date, Due Date, Returning Date, Total Number of Books, Fine Amount, Reader Barcode -> attributes
- System : Eliminated (too general)
- Reader card : Eliminated (out of scope)
- Interface / Page : Eliminated (too abstract)

S4: Consider quantitative relationship among classes

- A reader can have multiple borrowing receipts, but each borrowing receipt belongs to only one reader => Reader-BorrowingReceipt is a one-to-many relationship.
- A reader can have multiple returning receipt, and a returning receipt belongs to only one reader -> Reader - ReturningReceipt is a one-to-many relationship.
- A librarian can manage multiple borrowing receipts, but a single borrowing receipt can only be managed by one librarian -> User - BorrowingReceipt is a one-to-many relationship.
- A librarian can manage multiple returning receipts, but a returning receipt can only be managed by one

librarian -> User - ReturningReceipt is a one-to-many relationship.

- A book can be in multiple borrowing receipt, and a borrowing receipt can contain multiple books -> Book - BorrowingReceipt is an many-to-many relationship.

=> Proposed BorrowedBook: Each Book and BorrowingReceipt uniquely identifies a BorrowedBook, including the attributes: BorrowDate, DueDate, Price

=> Proposed ReturnedBook: Inherits from BorrowedBook, including the attributes: ReturnDate

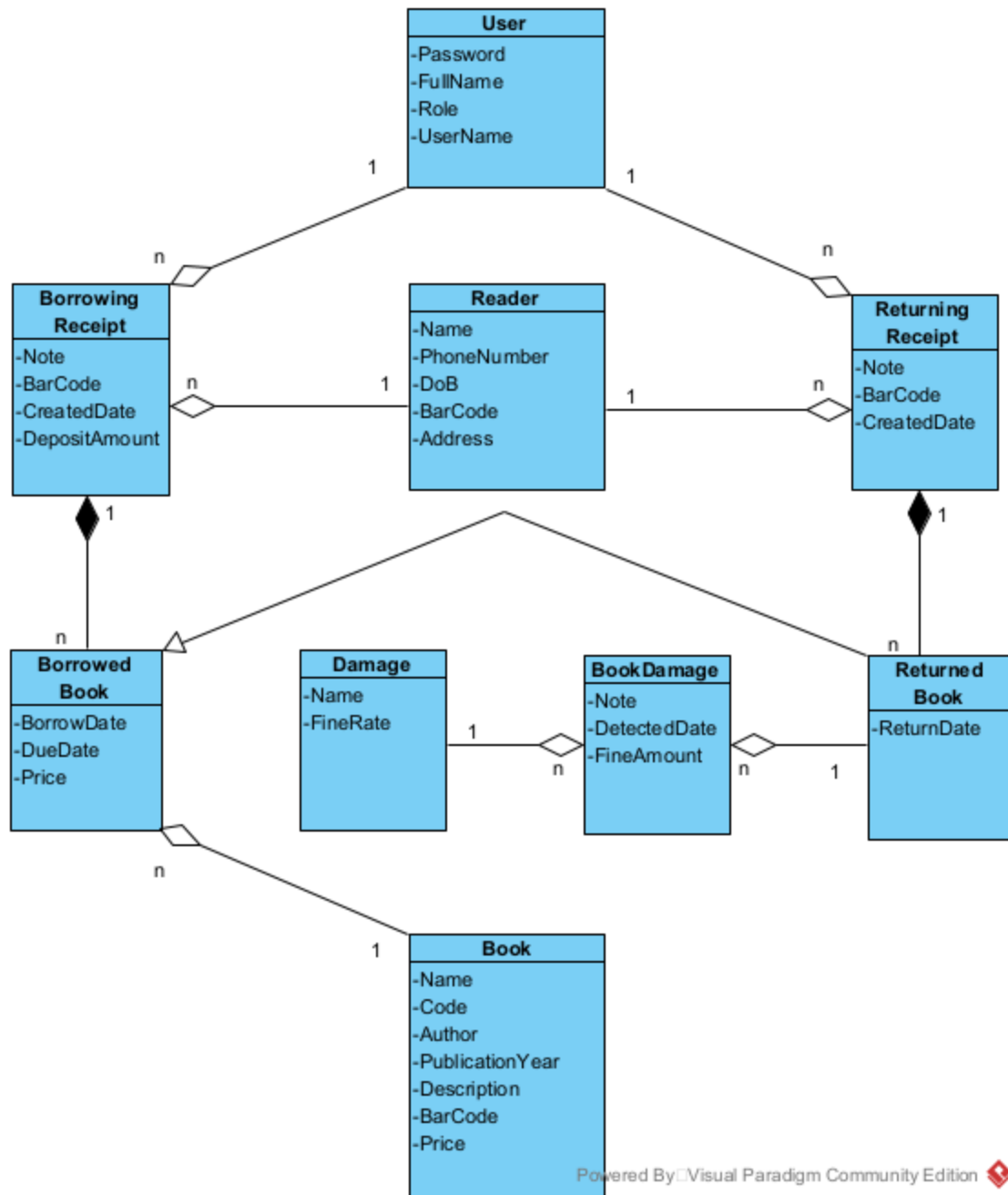
- A returning receipt has multiple returned books, and a returned book belongs to only one returning receipt -> The relationship between ReturningReceipt and ReturnedBook is an one-to-many relationship.

- A borrowing receipt has multiple borrowed books, and a borrowed book belongs to only one borrowing receipt -> The relationship between BorrowingReceipt and BorrowedBook is an one-to-many relationship.

- A returned book can have multiple existed damage, and a single damage can appear in multiple returned books -> ReturnedBook - Damage is an n-to-n relationship.

-> BookDamage: Each ReturnedBook and Damage uniquely identifies a BookDamage, including the attributes: Note, DetectedDate, FineAmount

S5 Add object relationship into diagram



4. Full classes diagram of the Analysis phase

I. Extraction

- Log in to the system → The login interface appears → Need a class: LoginView
 - + Username input field → inUsername
 - + Password input field → inPassword
 - + Login button → subLogin
- After entering the username and password and clicking login → the system checks whether the account is valid → need a function to verify login.
 - + Name: checkLogin()
 - + Input: Username, Password (User)
 - + Output: Boolean
 - + Actor Class: User

- Login successfully → display the main interface of the librarian staff → need class: LibrarianView

- + Button to switch to the reader search interface → subReturnBook
- + Cancel button → subCancel

- Click the return book option → display the reader search interface → need class: SearchReaderView

- + Input field → inSearchReader
- + Scan button → subScanReader
- + Search button → subSearchReader
- + List of readers that can be clicked to select → outsubListReader

- After entering the reader's name and clicking search, the system searches for readers whose names contain all or part of the entered keyword → need a function to search readers by name.

- + Name: searchReaderByName()
- + Input: a keyword
- + Output: List<Reader>
- + Actor Class: Reader

- Select scan reader, the system receives the reader code sent from the scanner and searches for the reader with the same code → need a function to search reader by code.

- + Name: searchReaderByBarCode()
- + Input: bar code
- + Output: an object of Reader
- + Actor Class: Reader

- Select a reader from the search results → the system displays the reader information interface → need a class: ReaderInformationView

- + Reader information list → outReaderInformation
- + List of borrowed books not yet returned → outListOfBookBorrowing
- + List of scanned books → outListOfScannedBooks
- + List of returned books that can be edited → outinListOfBookReturned
- + Back button → subBack
- + Scan book button → subScanBook
- + Confirm button → subNext

- To display the list of borrowed books in the reader information interface, need a function to get borrowed book information.

- + Name: getBorrowedBook()
- + Input: Reader
- + Output: an object of BorrowingReceipt
- + Actor Class: BorrowingReceipt

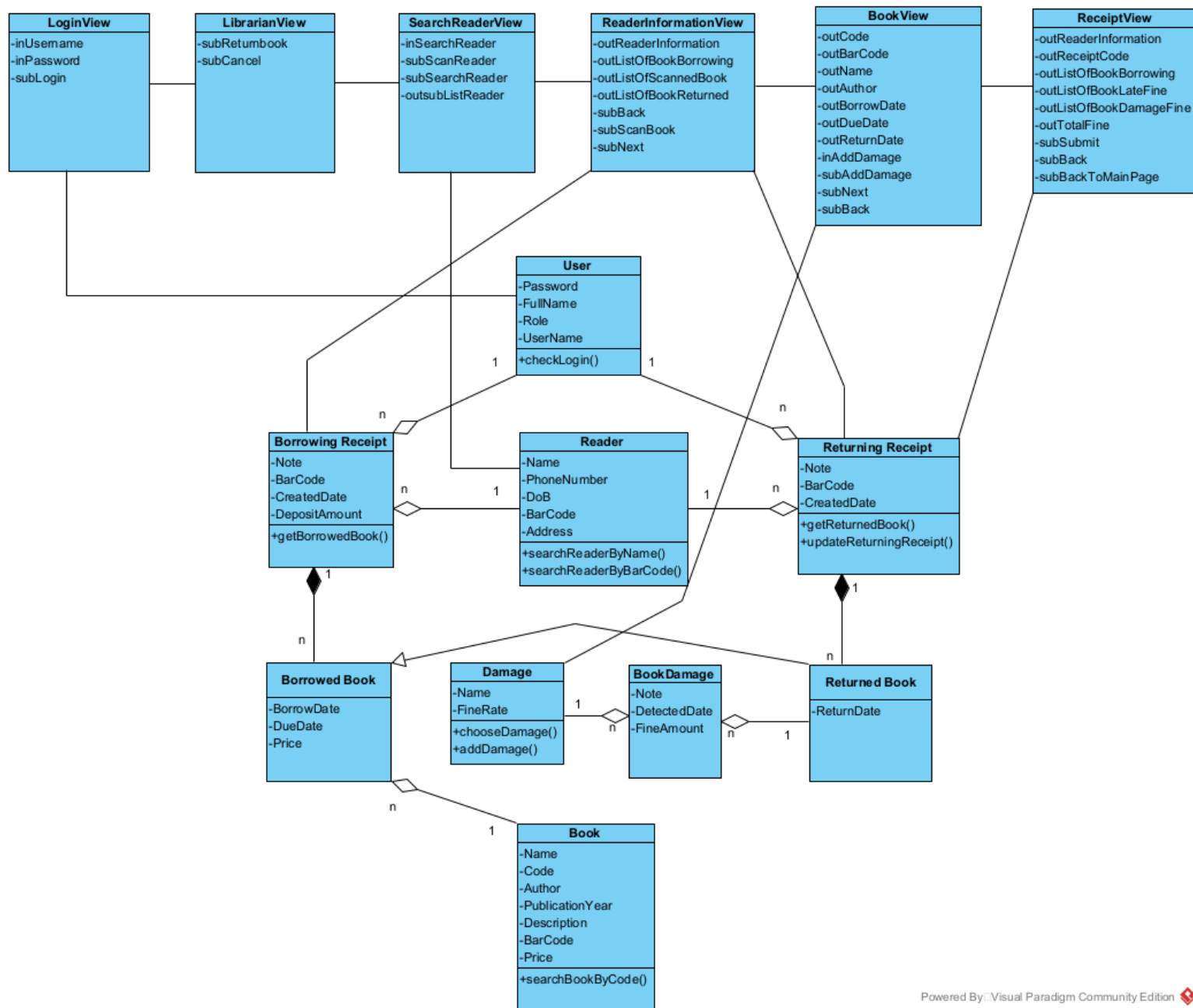
- To display the list of returned books in the reader information interface, need a function to get returned book information.

- + Name: getReturnedBook()

- + Input: Reader
 - + Output: List<ReturningReceipt>
 - + Actor Class: ReturningReceipt
- Click the scan book button, the system receives the book code from the scanner and searches for the book with the corresponding code → need a function to search for a book.
- + Name: searchBookByCode()
 - + Input: text
 - + Output: an object of Book
 - + Actor Class: Book
- Click on a row in the scanned books list, the system displays the book information interface → need a class: BookView
- + Code → outCode
 - + Barcode → outBarcode
 - + Name → outName
 - + Author → outAuthor
 - + Borrow date → outBorrowDate
 - + Due return date → outDueDate
 - + Return date → outReturnDate
 - + Add damage input field → inAddDamage
 - + Add damage button → subAddDamage
 - + Accept button → subNext
 - + Back button → subBack
- When the user selects a damage, the system needs a function to process the damage selection.
- + Name: chooseDamage()
 - + Input: an object of Damage
 - + Output: boolean
 - + Actor Class: Damage
- When the user clicks to add a new damage, the system needs a function to process adding the damage.
- + Name: addDamage()
 - + Input: an object of Damage
 - + Output: boolean
 - + Actor Class: Damage
- Click the Next button on the reader information interface → display the book return receipt interface → need a class: ReceiptView
- + Reader information → outReaderInformation
 - + Receipt code → outReceiptCode
 - + List of books not yet returned → outListOfBooksBorrowing
 - + List of late returned books → outListOfBooksLateFine
 - + List of damaged books → outListOfBooksDamageFine

- + List of deposit refunds → outListOfDeposit
 - + Total payment → outTotalFine
 - + Submit button → subSubmit
 - + Back button → subBack
 - + Back to main page → subBackToMainPage
- Click the Confirm button, the system updates the returned book information in the database → need a function:
- + Name: updateReturningReceipt()
 - + Input: an object of ReturningReceipt
 - + Output: none
 - + Actor Class: ReturningReceipt

II. Diagram



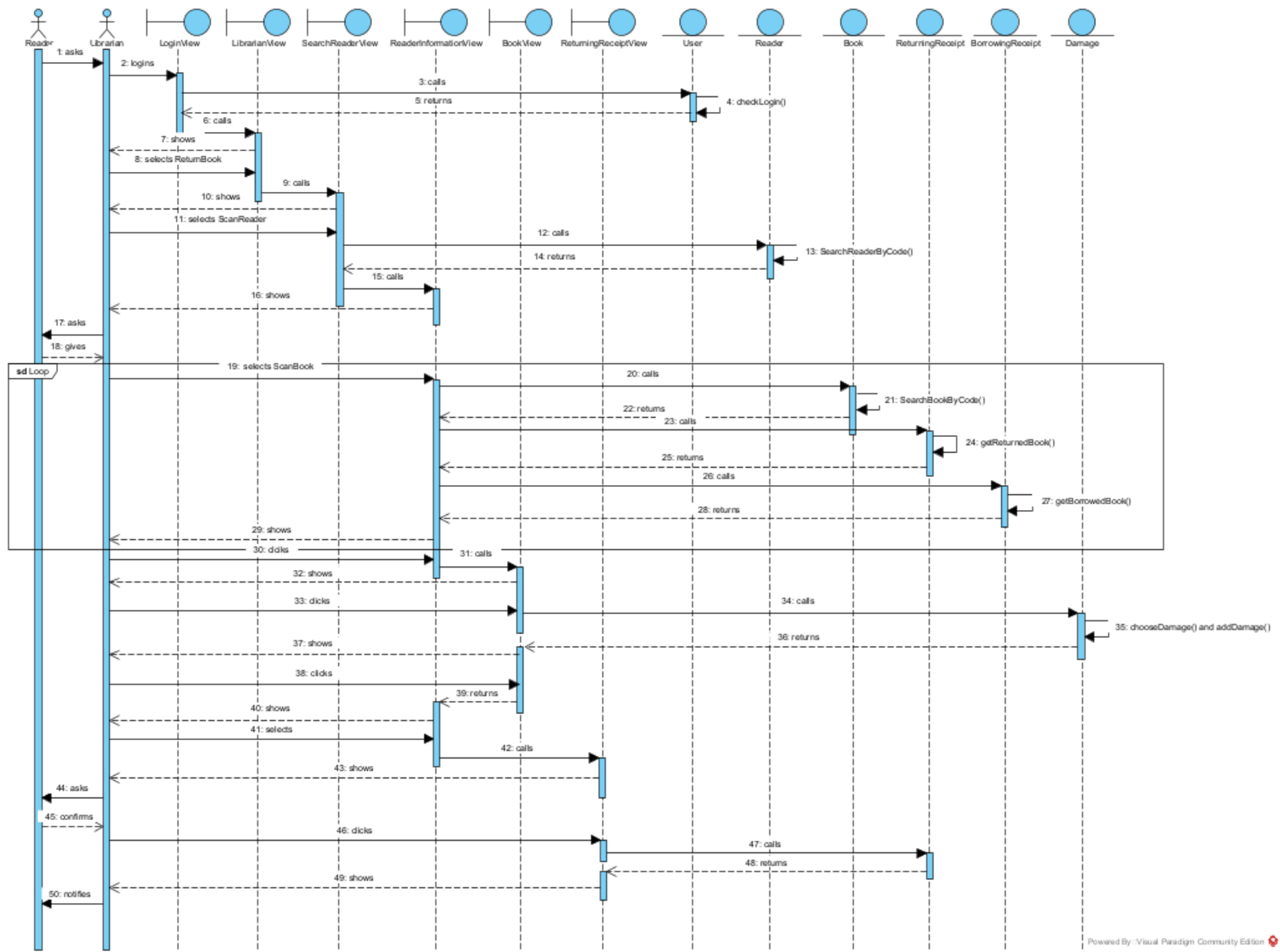
5. Sequence Diagram of the Analysis phase

I.Scenario V2

1. A reader asks a librarian to return books
2. Librarian logs in via the LoginView interface.
3. LoginView calls the User class.
4. The User class executes the checkLogin() function.
5. The User class returns the result to LoginView.
6. LoginView calls the LibrarianView class.
7. LibrarianView is displayed to the librarian.
8. Staff selects ReturnBook from the LibrarianView.
9. LibrarianView calls the SearchReaderView class.
10. SearchReaderView is displayed to the librarian.
11. Librarian clicks ScanReader on the SearchReaderView interface.
12. SearchReaderView calls the Reader class.
13. The Reader class executes the SearchReaderByCode() function.
14. The Reader class returns the result to SearchReaderView.
15. SearchReaderView calls the ReaderInformationView class.
16. ReaderInformationView is displayed to the librarian.
17. Librarian asks the reader which books they wish to return.
18. The reader hands the books over to the librarian. *(Repeat steps 19 to 29 until all books provided by the reader are scanned)*
19. Librarian selects ScanBook on the ReaderInformationView interface.
20. ReaderInformationView calls the Book class.
21. The Book class executes the SearchBookByCode() function.
22. The Book class returns the result to ReaderInformationView.
23. ReaderInformationView calls the ReturningReceipt class.
24. The ReturningReceipt class executes the getReturnedBook() function.
25. The ReturningReceipt class returns the result to ReaderInformationView.
26. ReaderInformationView calls the BorrowingReceipt class.
27. The BorrowingReceipt class executes the getBorrowedBook() function.
28. The BorrowingReceipt class returns the result to ReaderInformationView.
29. ReaderInformationView is displayed to the librarian.
30. Librarian clicks on each book in the scanned books list on ReaderInformationView
31. ReaderInformationView calls BookView class
32. BookView is displayed to the librarian
33. The librarian clicks Add button on BookView
34. BookView calls the Damage class
35. The Damage class executes the chooseDamage() and addDamage().
36. The Damage class returns the result to BookView
37. BookView is displayed to the librarian
38. The BookView clicks Back button on BookView
39. The BookView returns ReaderInformationView
40. The ReaderInformationView is displayed for librarian
41. Librarian clicks the Next button on the ReaderInformationView interface
42. ReaderInformationView calls the ReturningReceiptView class.
43. ReturningReceiptView is displayed to the librarian.
44. Librarian allows the reader to confirm the return receipt information.

45. The reader confirms the information is correct.
46. Librarian clicks the Submit button on the ReturningReceiptView interface.
47. ReturningReceiptView calls the ReturningReceipt class.
48. The ReturningReceipt class executes the updateReturningReceipt() function.
49. The ReturningReceipt class returns the result to ReturningReceiptView.
50. ReturningReceiptView displays a success message to the librarian.
51. Librarian notifies the reader that the process is successful.

II. Sequence diagram



6. Entity Class Designing

S1: Add id attribute into classes that are not inherited from others and add attribute type

=> User, Reader, Returning Receipt, Borrowing Receipt, Borrowed Book, Returned Book, Damage, Book
Damage, Book

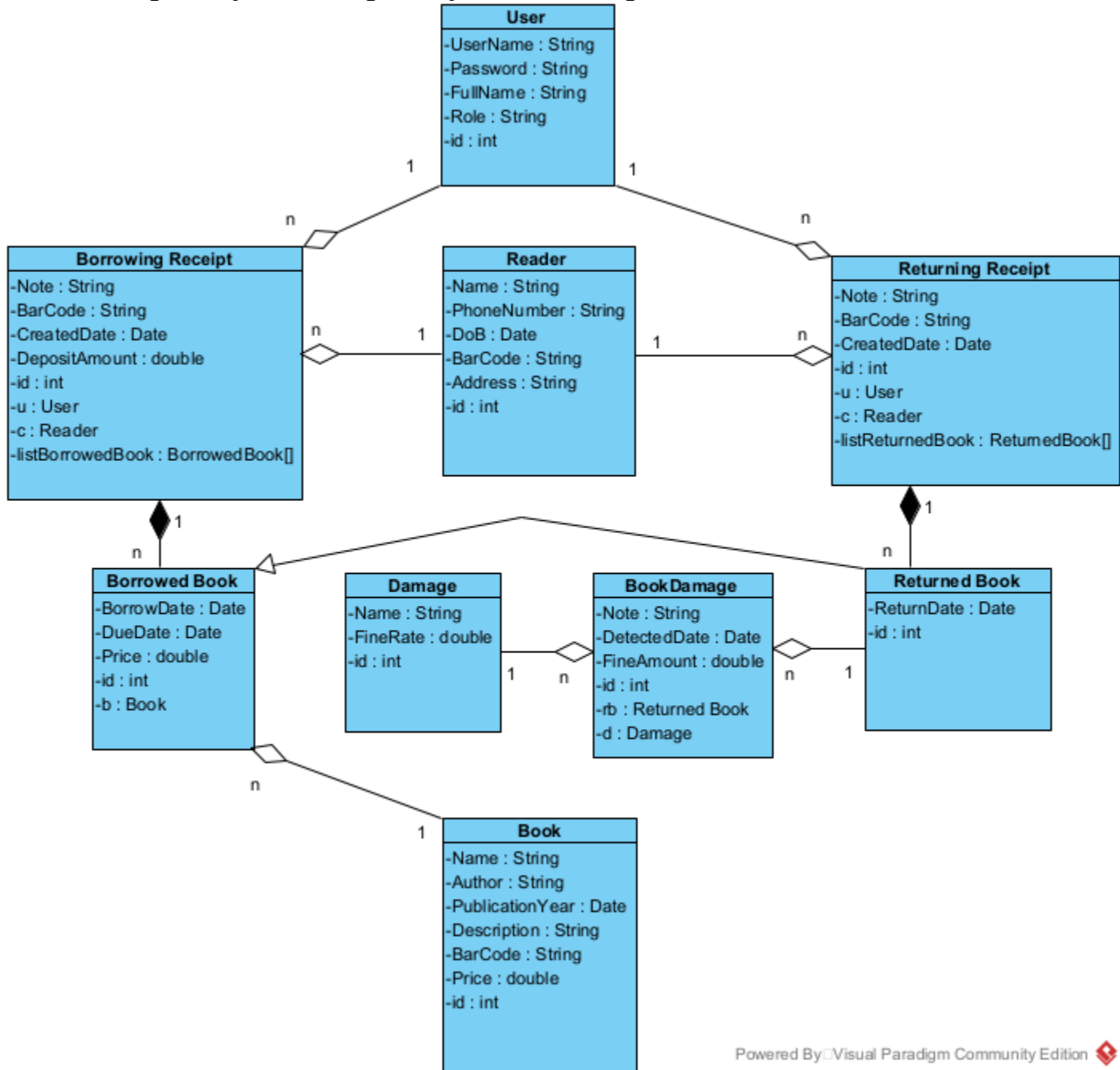
S2: Convert association relationships into aggregation/composition relationships

S3: Add object attribute for aggregation/compositon relationships

- User is included in BorrowingReceipt, 1 – n relationship => BorrowingReceipt has 1 User
- User is included in ReturningReceipt, 1 – n relationship => ReturningReceipt has 1 User
- Reader is included in BorrowingReceipt , 1 – n relationship => BorrowingReceipt has 1 Reader
- Reader is included in ReturningReceipt , 1 – n relationship => ReturningReceipt has 1 Reader
- BorrowedBook is included in BorrowingReceipt, n – 1 relationship => BorrowingReceipt has a list of BorrowedBook
- ReturnedBook is included in ReturningReceipt, n – 1 relationship => ReturningReceipt has a list of

ReturnedBook

- Damage is included in BookDamage, 1 – n relationship => BookDamage has 1 Damage
 - ReturnedBook is included in BookDamage, 1 – n relationship => BookDamage has 1 ReturnedBook
 - Book is included in BorrowedBook, 1 – n relationship => BorrowedBook has 1 Book
- => Borrowing Receipt, Returning Receipt, Book Damage, Borrowed Book, Returned Book.



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7. Database Designing

Input: entity class diagram of design

S1: Each entity class is converted into a table

User → tblUser

Reader → tblReader

BorrowingReceipt → tblBorrowingReceipt
ReturningReceipt → tblReturningReceipt
Damage → tblDamage
BookDamage → tblBookDamage
BorrowedBook → tblBorrowedBook
ReturnedBook → tblReturnedBook
Book → tblBook

S2: Convert non-object attributes of the entity class into attributes of the corresponding table.

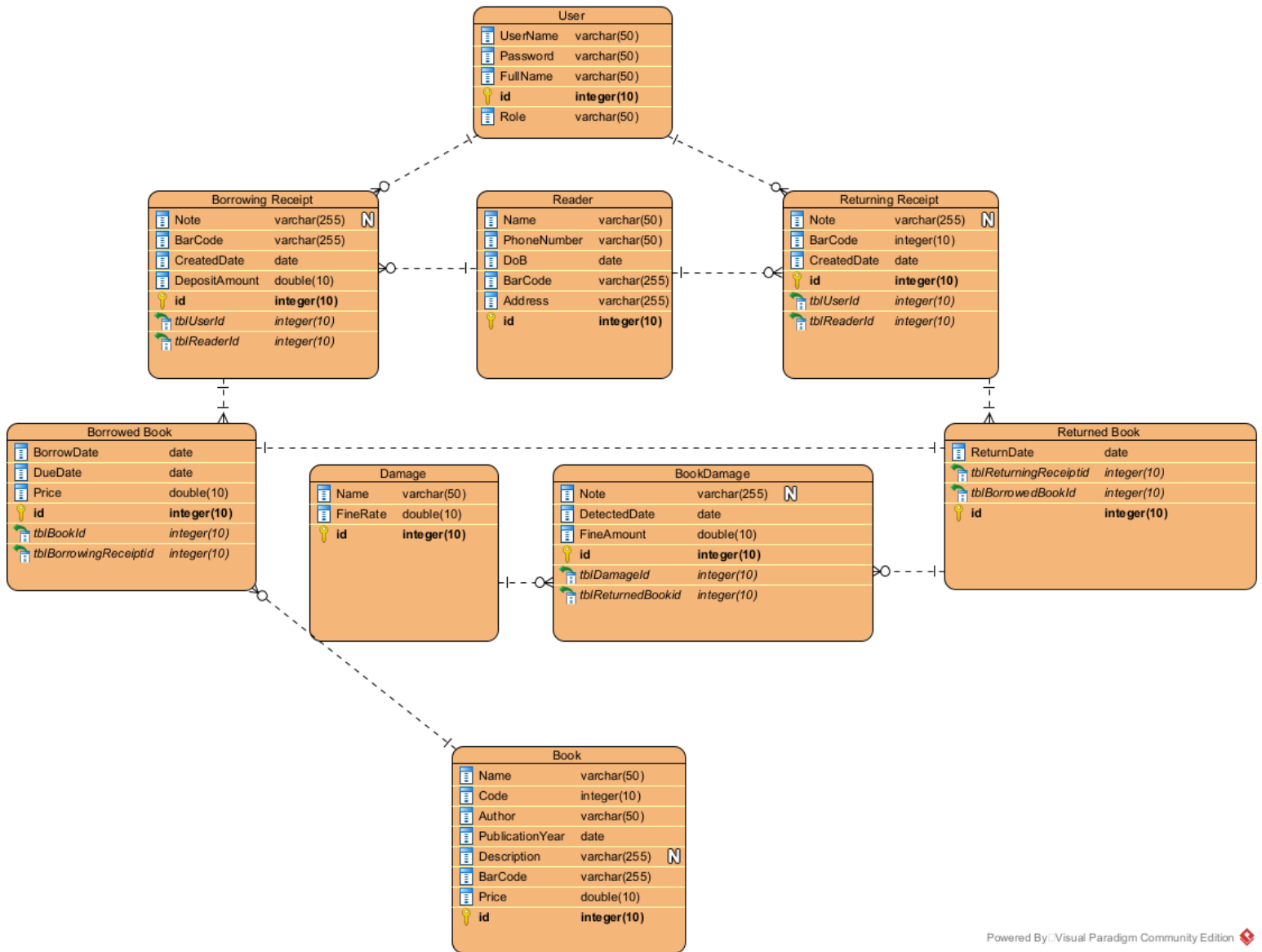
tblUser: : id, Username, Password, FullName, Role.
tblReader: Name, PhoneNumber, DateOfBirth, Barcode, Address, id.
tblReturningReceipt: Barcode, Note, CreatedDate, id.
tblBorrowingReceipt: Barcode, Note, CreatedDate, DepositAmount, id.
tblBorrowedBook: BorrowDate, DueDate, Price, id.
tblReturnedBook: ReturnDate, id
tblBook: Name, Code, Author, PublicationYear, Description, Barcode, Price, id.
tblDamage: Name, FineRate, id.
tblBookDamage: Note, DetectedDate, FineAmount, id.

S3: Quantative relationships among classes

1 tblUser - 0..n tblReturningReceipt
1 tblUser - 0..n tblBorrowingReceipt
1 tblReader - 0..n tblReturningReceipt
1 tblReader - 0..n tblBorrowingReceipt
1 tblBook - 0..n tblBorrowedBook
1 tblReturnedBook - 0..n tblBookDamage
1 tblBorrowedBook - 0..1 tblReturnedBook
1 tblDamage – 0..n tblBookDamage
1 tblReturningReceipt - 1..n tblReturnedBook
1 tblBorrowingReceipt - 1..n tblBorrowedBook

S4: Add primary key, foreign key.

1 tblUser - 0..n tblReturningReceipt -> tblReturningReceipt has a foreign key tblUserid
1 tblUser - 0..n tblBorrowingReceipt -> tblBorrowingReceipt has a foreign key tblUserid
1 tblReader - 0..n tblReturningReceipt -> tblReturningReceipt has a foreign key tblReaderid
1 tblReader - 0..n tblBorrowingReceipt -> tblBorrowingReceipt has a foreign key tblReaderid
1 tblBook - 0..n tblBorrowedBook -> tblBorrowedBook has a foreign key tblBookid
1 tblReturnedBook – 0..n tblBookDamage -> tblBookDamage has a foreign key tblReturnedBookid
1 tblDamage – 0..n tblBookDamage -> tblBookDamage has a foreign key tblDamageid
1 tblReturningReceipt - 1..n tblReturnedBook -> tblReturnedBook has a foreign key tblReturningReceiptid
1 tblBorrowingReceipt - 1..n tblBorrowedBook -> tblBorrowedBook has a foreign key tblBorrowingReceiptid
1 tblBorrowedBook – 1 tblReturnedBook -> tblReturnedBook has a foreign key tblBorrowedBookid



8.Interface Designing and Full Classes Diagram of the Design phase

I. Interface designing

Librarian Login

LOGIN

Username:

Password:

Librarian Home

LIBRARIAN HOME

Librarian FullName

Search Reader

SEARCH READER

Search Reader

Reader ID	Name	DateOfBirth	Address	Number	Barcode
001	Nguyen Van A	2000-01-01	Ha Noi	0123456789	0123456788
002	Tran Thi B	1999-05-10	Nam Dinh	0987654321	0123456789

READER DETAIL

Reader Information

Reader ID	Name	DoB	Address	Phone number	Bar Code
001	B	01/01/2026	Ha Noi	0123456789	0123456789

List of borrowed books that have not been returned

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover Price
1	00001	12345	Harry Potter	J.K Rowling	11/03/2026	11/04/2026	10000vnd
2	00002	12346	Diary of a Cricket	To Hoai	11/03/2026	11/04/2026	20000vnd

List of borrowed books that have been returned

Order	Book Code	Bar Code	Name	Author	Cover Price	Due Date	Returning Date	Fine Amount
1	00003	12347	Tam Cam	Unknown	30000vnd	01/04/2026	20/03/2026	0vnd
2	00004	12348	The lotus shoes	Jane Yang	40000vnd	01/04/2026	21/03/2026	40000vnd

Scan book

Back

Next

BOOK RETURN

Reader Information

Reader ID	Name	DoB	Address	Phone number	Bar Code
001	B	01/01/2026	Ha Noi	0123456789	0123456789

List of borrowed books that have not been returned

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover Price
1	00001	12345	Harry Potter	J.K Rowling	11/03/2026	11/04/2026	10000vnd

List of scanned books

Order	Book Code	Bar Code	Name	Author	Due Date	Returning Date	Cover Price	Damage Statu...	Damage Statu...
1	00002	12346	Diary of a Cricket	To Hoai	11/04/2026	12/04/2026	20000vnd	OK	TORN

List of borrowed books that have been returned

Order	Book Code	Bar Code	Name	Author	Cover Price	Due Date	Returning Date	Fine Amount
1	00003	12347	Tam Cam	Unknown	30000vnd	01/04/2026	20/03/2026	0vnd
2	00004	12348	The lotus shoes	Jane Yang	40000vnd	01/04/2026	21/03/2026	40000vnd

Scan book

Back

Next

Select Damage State

BOOK DAMAGE SELECTION

- Book Code: 000002

- BarCode: 12346

- Name: Diary of a Cricket

- Author: Tô Hoài

- Borrowing Date: 11/03/2026

- Due Date: 11/04/2026

- Returning Date: 12/04/2026

- Cover Price: 20000vnd

- Type of Damage:

Type	Fine Amount (Based on Cover...	Select
Torn	100%	☐
Graffiti	50%	☐
Mold	100%	☐
Crumpled	100%	☐
Bent	80%	☐

Add

Back

Accept

Book Return Invoice

RETURNING RECEIPT

Reader Information

Reader ID	Name	DoB	Address	Phone number	Bar Code	Script Code
001	B	01/01/2026	Ha Noi	0123456789	0123456789	1234561234

List of borrowed books that have not been returned

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover Price
1	00001	12345	Harry Potter	J.K Rowling	11/03/2026	11/04/2026	10000vnd

List of books fined for late returning

Order	Book Code	Bar Code	Name	Author	Borrowing D...	Due Date	Returning D...	Cover Price	Fine Amount...
1	00002	12346	Diary of a Cri...	Tô Hoài	11/03/2026	11/04/2026	12/04/2026	20000vnd	4000vnd

List of books fined for returning damaged book

Order	Book Code	Bar Code	Name	Author	Cover Price	Damaged Status...	Fine Amount
1	00002	12346	Diary of a Cricket	Tô Hoài	20000vnd	TORN	20000vnd

Total Fine: 24000vnd

Deposit: 20000vnd

Submit

Back to Main Page

Back

II. Full classes diagram of the Design phase

- Design based on MVC model:

View classes:

- + **LoginFrm**: A login interface for librarians, requires 1 text field for entering username, 1 password field for entering password, 1 Login button for authentication, and 1 Cancel button to exit the program.
- + **LibrarianHomeFrm**: The main interface for librarians after successful login, requires 1 label displaying the librarian's name ,1 Return Book button to access the book return function, and 1 Cancel button to exit.
- + **SearchReaderFrm**: An interface for searching reader information, requires 1 text field for entering search keywords, 1 Search button to execute the search, 1 Scan Card button to scan the reader's card, and 1 table displaying the list of readers
- + **ReaderDetailFrm** A detailed reader information interface, requires 1 table displaying reader information , 1 table displaying the list of books borrowed but not yet returned, 1 table displaying the list of returned books, 1 Scan book button to scan books, 1 Back button to return, and 1

Next button to proceed.

+ **BookReturnFrm**: A book return interface, requires 1 table displaying reader information , 1 table displaying the list of borrowed but unreturned books, 1 table displaying the list of scanned books with the Damage Status Now column as a button to select damage type, 1 table displaying the list of returned books, 1 Scan book button, 1 Back button, and 1 Next button.

+ **BookDamageFrm**: A book damage status selection interface, requires 8 labels displaying book information, 1 Type of Damage label, 1 table displaying the list of damage types, 1 text field for entering a new damage type name, 1 Add button, 1 Back button, and 1 Accept button.

+ **ReturningReceiptFrm**: A book return receipt interface, requires 1 table displaying reader information , 1 table displaying the list of borrowed but unreturned books, 1 table displaying the list of books fined for late return , 1 table displaying the list of books fined for damage, 1 label displaying the total amount, 1 Submit button to confirm, 1 Back to Main Page button to return to the main page, and 1 Back button to go back.

Control Classes (DAO)

+ **Base DAO Class**

BaseDAO: provides 1 method to establish database connection for all DAO classes that inherit from it in the system.

+ **UserDAO**

Requires 1 method to check login information.

Method name: checkLogin

Input: u: User (username, password)

Output: boolean

DAO: UserDAO

+ **ReaderDAO**

Requires method to search readers by name and by code.

Method name: searchReaderByName

Input: S: String

Output: Reader[]

DAO: ReaderDAO

Method name: searchReaderByBarCode

Input: S: String

Output: Reader[]

DAO: ReaderDAO

+ **BookDAO**

Requires method to get book information

Method name: searchBookByCode

Input: S: String

Output: Book

DAO: BookDAO

+ **BorrowingReceiptDAO**

Requires methods to get borrowed books.

Method name: getBorrowedBook

Input: r: Reader

Output: BorrowingReceipt

DAO: BorrowingReceiptDAO

+ DamageDAO

Requires methods to add damage and choose damage

Method name: chooseDamage

Input: d: Damage

Output: Boolean

DAO: DamageDAO

Method name: addDamage

Input: d: Damage

Output: Boolean

DAO: DamageDAO

+ ReturningReceiptDAO

Requires methods to get returnedBook and create Returning Receipt

Method name: getReturnedBook

Input: r:Reader

Output: ReturningReceipt[]

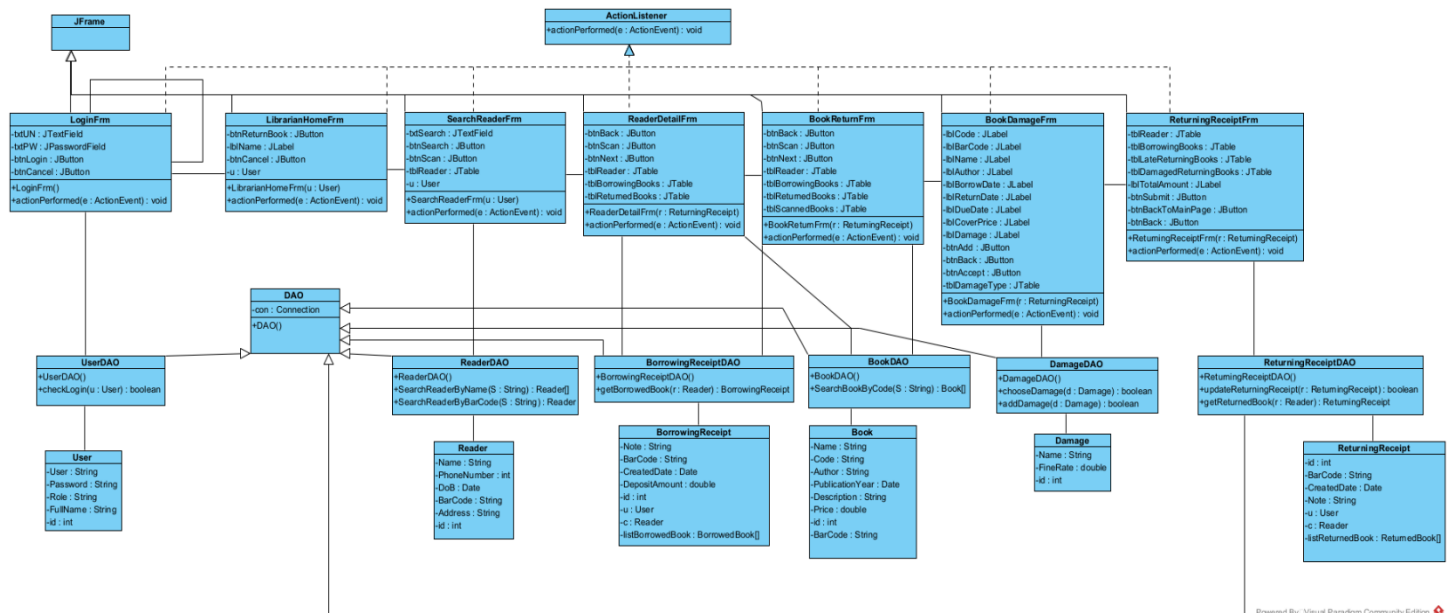
DAO: ReturningReceiptDAO

Method name: updateReturningReceipt

Input: r: ReturningReceipt

Output: boolean

DAO: ReturningReceiptDAO

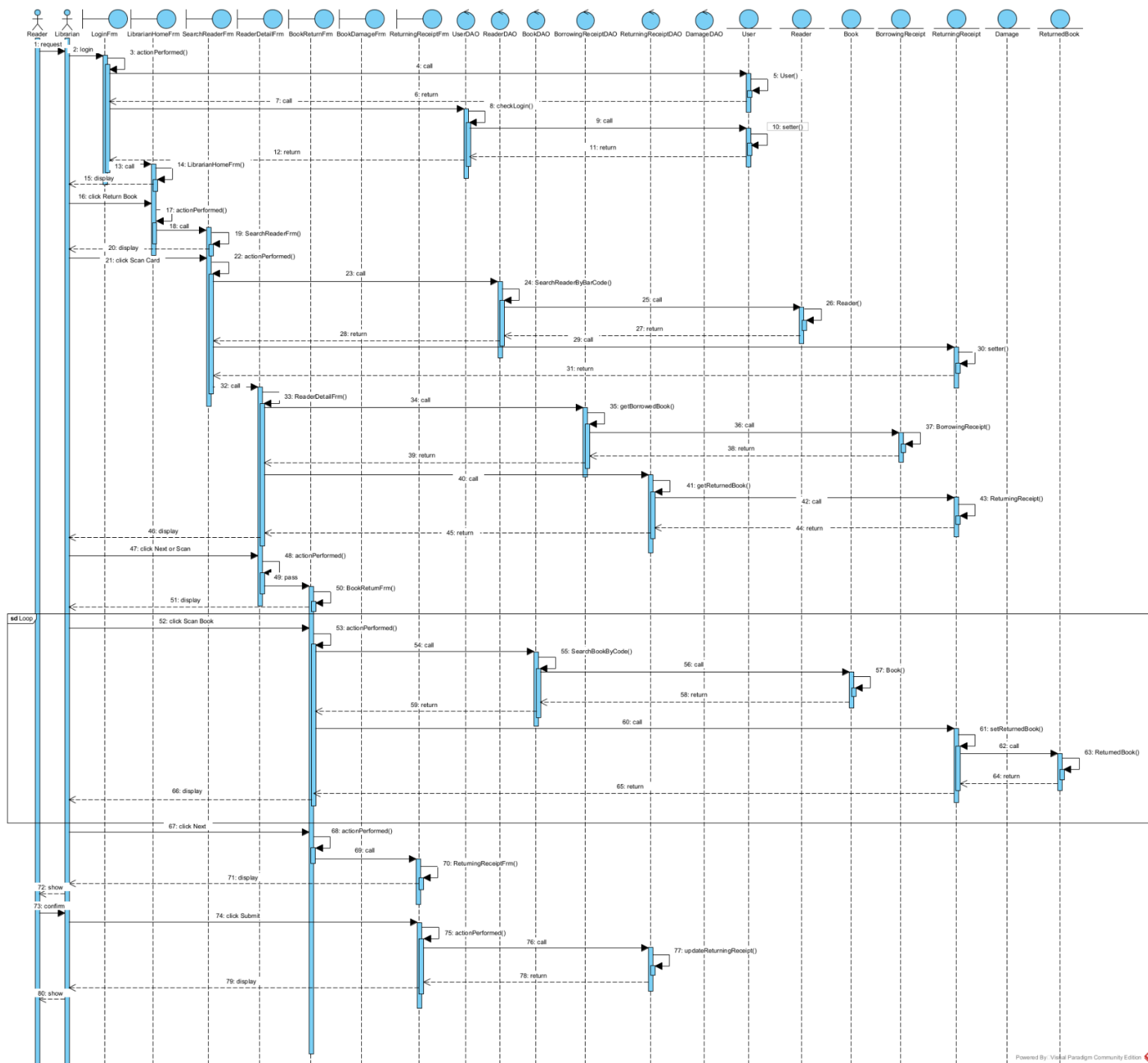


9. Sequence Diagram of the Design phase

- Scenario v3

1. The reader requests support from the librarian to return borrowed books.
2. The librarian enters username and password, then clicks the Login button on LoginFrm.
3. The actionPerformed() method of LoginFrm is called.
4. The method actionPerformed() calls User to create an User object
5. The class User packs the information into an User object
6. The class User returns User object to the method actionPerformed()
7. The method actionPerformed() calls method checkLogin() of the class UserDAO
8. The method checkLogin() checks the login information
9. The method checkLogin() calls the class User to set 2 attributes fullname and role.
10. The class User calls its method setFullName(), setRole()
11. The class User returns the User object to the method checkLogin()
12. The method checkLogin() returns the results to the actionPerformed()
13. The method actionPerformed() calls the class LibrarianHomeFrm
14. The constructor LibrarianHomeFrm() is called
15. The interface LibrarianHomeFrm is display to the librarian
16. The manager clicks on the Return Book button
17. The actionPerformed() method of LibrarianHomeFrm is called
18. The method actionPerformed() calls the class SearchReaderFrm
19. The constructor SearchReaderFrm() is called
20. The interface SearchReaderFrm is display to the librarian
21. The librarian clicks Scan Card button
22. The method actionPerformed() of SearchReaderFrm is called
23. The method actionPerformed() calls SearchReaderByBarCode() of ReaderDAO
24. The method SearchReaderByBarCode() search readers by barcode
25. The method SearchReaderByBarCode() calls Reader class to create Reader object
26. The Reader class packs information into Reader object
27. The Reader class returns Reader object to the method SearchReaderByBarCode ()
28. The method SearchReaderByBarCode() returns the results to the method actionPerformed() of the class SearchReaderFrm
29. The actionPerformed() method calls the ReturningReceipt class to create a ReturningReceipt object
30. The ReturningReceipt class packs the Reader information into the ReturningReceipt object.
31. The ReturningReceipt class returns the ReturningReceipt object to the actionPerformed() method.
32. The actionPerformed() method calls the ReaderDetailFrm class.
33. The constructor of ReaderDetailFrm is called.
34. The constructor of ReaderDetailFrm calls the getBorrowedBook() method of the BorrowingReceiptDAO class
35. The getBorrowedBook() method searches BorrowingReceipt by reader ID
36. The getBorrowedBook() method calls the BorrowingReceipt class to create a BorrowingReceipt object
37. The BorrowingReceipt class packs the information into the BorrowingReceipt object
38. The BorrowingReceipt class returns the BorrowingReceipt object to the getBorrowedBook() method
39. The getBorrowedBook() method returns the result to the constructor of ReaderDetailFrm
40. The constructor of ReaderDetailFrm calls the getReturnedBook() method of the ReturningReceiptDAO class
41. The getReturnedBook() method searches ReturningReceipt by reader ID.

42. The `getReturnedBook()` method calls the `ReturningReceipt` class to create `ReturningReceipt` objects.
43. The `ReturningReceipt` class packs the information into the `ReturningReceipt` objects.
44. The `ReturningReceipt` class returns the `ReturningReceipt` objects to the `getReturnedBook()` method.
45. The `getReturnedBook()` method returns the result to the constructor of `ReaderDetailFrm`.
46. The `ReaderDetailFrm` interface is displayed to the librarian
47. The librarian clicks the Next or Scan Book button.
48. The `actionPerformed()` method of `ReaderDetailFrm` is called.
49. The `actionPerformed()` method passes Reader information, borrowed books list, returned books list and `ReturningReceipt` object to `BookReturnFrm`.
50. The constructor `BookReturnFrm` is called.
51. The `BookReturnFrm` interface is displayed to the librarian.
- (Repeat steps 52 to 66 as needed)
52. The librarian clicks the Scan Book button.
53. The `actionPerformed()` method of the `BookReturnFrm` class is called
54. The `actionPerformed()` method calls the `SearchBookByCode()` method of the `BookDAO` class.
55. The `SearchBookByCode()` method searches for a book by code.
56. The `SearchBookByCode()` method calls the `Book` class to create a `Book` object.
57. The `Book` class packs the information into the `Book` object.
58. The `Book` class returns the result to the `SearchBookByCode()` method.
59. The `SearchBookByCode()` method returns the result to the `actionPerformed` method of `BookReturnFrm`.
60. The `actionPerformed()` method calls the `ReturningReceipt` class.
61. The `setReturnedBook()` method of the `ReturningReceipt` class is called.
62. The `setReturnedBook()` method calls the `ReturnedBook` class.
63. The `ReturnedBook` class packs the information into the `ReturnedBook` object.
64. The `ReturnedBook` class returns the result to the `ReturningReceipt` class.
65. The `ReturningReceipt` class returns the result to the `BookReturnFrm` class.
66. The `BookReturnFrm` interface is displayed to the librarian.
67. The librarian clicks the Next button.
68. The `actionPerformed()` method of the `BookReturnFrm` class is called.
69. The `actionPerformed()` method calls the `ReturningReceiptFrm` class.
70. The constructor `ReturningReceiptFrm()` is called.
71. The `ReturningReceiptFrm` interface is displayed to the librarian
72. The librarian lets the reader check the returning receipt information.
73. The reader confirms that the information is correct.
74. The librarian clicks the Submit button.
75. The `actionPerformed()` method of the `ReturningReceiptFrm` class is called.
76. The `actionPerformed()` method calls the `updateReturningReceipt()` method of the `ReturningReceiptDAO` class.
77. The `updateReturningReceipt()` method saves the `ReturningReceipt` into the database.
78. The `ReturningReceipt` class returns the result to the `actionPerformed()` method.
79. The `actionPerformed()` method displays a success notification to the librarian.
80. The librarian informs the reader that the return process is successful and gives the receipt to the reader.



10. Blackbox test plan and Blackbox test case

Blackbox test plan

Order	Module	Testcase
1	Returning Book	Add returning receipt once, the reader exists in the list of readers, the returning books exist in the list of unreturned books
2		Add returning receipt once, the reader doesn't exist in the list of readers, the returning books exist in the list of unreturned books

3		Add returning receipt once, the reader exists in the list of readers, the returning books don't exist in the list of unreturned books
4		Add returning receipt once, the reader doesn't exist in the list of readers, the returning books don't exist in the list of unreturned books
5		Add the same returning receipt two consecutive times, the reader exists in the list of readers, the returning books exist in the list of unreturned books

=> **Blackbox test case (standard) with full test data:** Add returning receipt once, the reader exists in the list of readers, the returning books exist in the list of unreturned books

- The database before testing:

tblUser

id	Username	Password	FullName	Role
1	a	a@123	Nguyen Van A	Librarian

tblBook

id	Name	Code	Author	Publication Year	Description	Barcode	Price
1	Harry Potter	00001	J.K Rowling	01/01/2000	Good	12345	10000
2	Diary of a cricket	00002	Tô Hoài	01/01/2000	Dramatic	12346	20000
3	Tâm Cám	00003	Unknown	01/01/2000	Marvelous	12347	30000
4	The lotus shoes	00004	Jane Yang	01/01/2000	Incredible	12348	40000

tblBorrowedBook

id	BorrowDate	DueDate	Price	tblBookid	tblBorrowingReceiptid
1	01/04/2026	01/05/2026	10000	1	1
2	01/04/2026	01/05/2026	20000	2	1
3	15/04/2026	15/05/2026	30000	3	2
4	15/04/2026	15/05/2026	40000	4	2

tblReturnedBook

id	ReturnDate	tblReturningReceiptid	tblBorrowedBookid
1	08/05/2026	1	2
2	08/05/2026	1	3

tblReturningReceipt

id	Barcode	Note	CreatedDate	tblReaderid	tblUserid
1	12345	None	08/05/2026	1	1

tblBorrowingReceipt

id	Barcode	Note	CreatedDate	DepositAmount	tblReaderid	tblUserid
----	---------	------	-------------	---------------	-------------	-----------

1	23456	None	01/04/2026	30000	1	1
2	23457	None	15/04/2026	70000	1	1

tblReader

id	Name	DateOfBirth	Address	PhoneNumber	Barcode
1	B	01/01/1997	Hà Nội	123456789	1111111111
2	Bình	28/02/2005	Hà Nội	123123123	1111111112
3	xxx	11/11/2011	Hà Nội	123123456	1111111113

tblDamage

id	Name	FineRate
1	Torn	90
2	Stain	50
3	Graffiti	50

tblBookDamage

id	Note	DetectedDate	FineAmount	tblDamageid	tblReturnedBookid
1	Stain Damage	08/05/2026	15000	2	2
2	Graffiti Damage	08/05/2026	15000	3	2

- The testing scenario and expected results:

Scenario	Expected Results
1.Start the application	The Login interface appeared with: two text fields for entering username, password, a login button and a Cancel button. username password login cancel
2. Enter username=a, password=a@123 Click login.	The Librarian Home interface appeared with these two buttons: a Return Book button and a Cancel button Return Book Cancel
3. Click on the Return Book button	The Search Reader interface appeared with: a text field for entering keyword, a Search button and a Scan Card button. Search Reader

	Search Scan Card																																																															
4. Enter keyword = b and click on the Search button	List of readers appeared in the Search Reader interface. There are two readers in the results: Search Reader b Search Scan Card <table><tr><td>Reader ID</td><td>Name</td><td>DoB</td><td>Address</td><td>Phone number</td><td>Bar Code</td></tr><tr><td>001</td><td>B</td><td>01/01/1997</td><td>Hà Nội</td><td>123456789</td><td>1111111111</td></tr><tr><td>002</td><td>Bình</td><td>28/02/2005</td><td>Hà Nội</td><td>123123123</td><td>1111111112</td></tr></table>	Reader ID	Name	DoB	Address	Phone number	Bar Code	001	B	01/01/1997	Hà Nội	123456789	1111111111	002	Bình	28/02/2005	Hà Nội	123123123	1111111112																																													
Reader ID	Name	DoB	Address	Phone number	Bar Code																																																											
001	B	01/01/1997	Hà Nội	123456789	1111111111																																																											
002	Bình	28/02/2005	Hà Nội	123123123	1111111112																																																											
5. Click on the row of the reader B	The Reader Detail interface appeared with these three buttons: a Scan Book button, a Back button, a Next button. - Reader Information <table><tr><td>Reader ID</td><td>Name</td><td>DoB</td><td>Address</td><td>Phone number</td><td>Bar Code</td></tr><tr><td>001</td><td>B</td><td>01/01/1997</td><td>Hà Nội</td><td>123456789</td><td>1111111111</td></tr></table> - List of borrowed books that have not been returned <table><tr><td>Order</td><td>Book Code</td><td>Bar Code</td><td>Name</td><td>Authors</td><td>Borrowing Date</td><td>Due Date</td><td>Cover price</td></tr><tr><td>1</td><td>00001</td><td>12345</td><td>Harry Potter</td><td>J.K Rowling</td><td>01/04/2026</td><td>01/05/2026</td><td>10000vnd</td></tr><tr><td>2</td><td>00004</td><td>12348</td><td>The lotus shoes</td><td>Jane Yang</td><td>15/04/2026</td><td>15/05/2026</td><td>40000vnd</td></tr></table> The total books:2 - List of borrowed books that have been returned <table><tr><td>Order</td><td>Book Code</td><td>Bar Code</td><td>Name</td><td>Author</td><td>Cover price</td><td>Due Date</td><td>Returning Date</td><td>Total Fine Amount</td></tr><tr><td>1</td><td>00002</td><td>12346</td><td>Diary of a cricket</td><td>Tô Hoài</td><td>20000vnd</td><td>01/05/2026</td><td>08/05/2026</td><td>4000vnd</td></tr><tr><td>2</td><td>00003</td><td>12347</td><td>Tám căm</td><td>Unknown</td><td>30000vnd</td><td>15/05/2026</td><td>08/05/2026</td><td>30000vnd</td></tr></table> The total of returned books2 Scan book Back Next	Reader ID	Name	DoB	Address	Phone number	Bar Code	001	B	01/01/1997	Hà Nội	123456789	1111111111	Order	Book Code	Bar Code	Name	Authors	Borrowing Date	Due Date	Cover price	1	00001	12345	Harry Potter	J.K Rowling	01/04/2026	01/05/2026	10000vnd	2	00004	12348	The lotus shoes	Jane Yang	15/04/2026	15/05/2026	40000vnd	Order	Book Code	Bar Code	Name	Author	Cover price	Due Date	Returning Date	Total Fine Amount	1	00002	12346	Diary of a cricket	Tô Hoài	20000vnd	01/05/2026	08/05/2026	4000vnd	2	00003	12347	Tám căm	Unknown	30000vnd	15/05/2026	08/05/2026	30000vnd
Reader ID	Name	DoB	Address	Phone number	Bar Code																																																											
001	B	01/01/1997	Hà Nội	123456789	1111111111																																																											
Order	Book Code	Bar Code	Name	Authors	Borrowing Date	Due Date	Cover price																																																									
1	00001	12345	Harry Potter	J.K Rowling	01/04/2026	01/05/2026	10000vnd																																																									
2	00004	12348	The lotus shoes	Jane Yang	15/04/2026	15/05/2026	40000vnd																																																									
Order	Book Code	Bar Code	Name	Author	Cover price	Due Date	Returning Date	Total Fine Amount																																																								
1	00002	12346	Diary of a cricket	Tô Hoài	20000vnd	01/05/2026	08/05/2026	4000vnd																																																								
2	00003	12347	Tám căm	Unknown	30000vnd	15/05/2026	08/05/2026	30000vnd																																																								
6. Click on the	The Book Return interface appeared with these three buttons: a Scan Book button, a																																																															

Scan Book button and use Scanner to scan the book with BarCode “12348”

Back button, a Next button.

- Reader Information

Reader ID	Name	DoB	Address	Phone number	Bar Code
001	B	01/01/1997	Hà Nội	123456789	1111111111

- List of borrowed books that have not been returned

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover price
1	00001	12345	Harry Potter	J.K Rowling	01/04/2026	01/05/2026	10000vnd
The total books:						2	

- List of scanned books

Order	Book Code	Bar Code	Name	Author	Due Date	Returning Date	Cover price	Damage Status Now
1	00004	12348	Harry Potter	J.K Rowling	15/05/2026	09/05/2026	40000vnd	OK
The total books:								1

- List of borrowed books that have been returned

Order	Book Code	Bar Code	Name	Author	Cover price	Due Date	Returning Date	Total Fine Amount
1	00002	12346	Diary of a cricket	Tô Hoài	20000vnd	01/05/2026	08/05/2026	4000vnd
2	00003	12347	Tấm cám	Unknown	30000vnd	15/05/2026	08/05/2026	30000vnd
The total of returned books						2		

7. Click on the Next button

The Returning Receipt interface appeared with these three buttons: a Submit button, a Back to Main page button, a Back button.

- Reader Information:

Reader ID	Name	DoB	Address	Phone number	Bar Code	ScriptCode
001	B	01/01/1997	Hà Nội	123456789	1111111111	12346

- List of borrowed books that have not been returned

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Cover price
1	00001	12345	Harry Potter	J.K Rowling	01/04/2026	01/05/2026	10000vnd
The total books:						1	

- List of books fined for late returning

Order	Book Code	Bar Code	Name	Author	Borrowing Date	Due Date	Returning Date	Cover price	Fine Amount (20%)
The total					0vnd				

- List of books fined for returning damaged book

Order	Book Code	Bar Code	Name	Author	Cover price	Damaged Status Now	Fine Amount
The total					0vnd		

TOTAL FINE: 0vnd DEPOSIT: 40000vnd

8. Click on the Submit button

The notification appeared with the message “Returning books successfully”

- The database after testing:

Only 2 tables change:

tblReturnedBook

id	ReturnDate	tblReturningReceiptid	tblBorrowedBookid
1	08/05/2026	1	2
2	08/05/2026	1	3
3	09/05/2026	2	4

tblReturningReceipt

id	Barcode	Note	CreatedDate	tblReaderid	tblUserid
1	12345	None	08/05/2026	1	1
2	12346	None	09/05/2026	1	1